

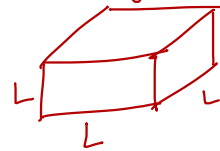
More about the Fermi gas

When the Fermi distribution function looks like a step function, the Fermi gas is said to be **degenerate**. As the temperature is raised, it becomes more and more non-degenerate and the distribution resembles the Maxwell-Boltzmann distribution.

The value of the chemical potential μ at $T=0$ is known as the **Fermi energy**, ϵ_F .

For free particles in volume V , under periodic boundary conditions, we have plane wave eigenstates.

$$\psi_{\vec{k}}(\vec{r}) = \frac{1}{\sqrt{V}} e^{i\vec{k} \cdot \vec{r}}$$



Imposing periodic boundary condition along the x-direction,

$$\psi_{\vec{k}}(x+L, y, z) = \psi_{\vec{k}}(x, y, z) \Rightarrow e^{ik_x L} = 1 \Rightarrow k_x = \frac{2\pi n_x}{L},$$

where n_x is an integer.

The same is true for the y and z directions

$$\Rightarrow \vec{k} = \frac{2\pi}{L} (n_x \hat{x} + n_y \hat{y} + n_z \hat{z}) \quad n_x, n_y, n_z \in \mathbb{Z}$$

$$\therefore \sum_{\vec{k}} f(\vec{k}) = \sum_{n_x=-\infty}^{\infty} \sum_{n_y=-\infty}^{\infty} \sum_{n_z=-\infty}^{\infty} f(\vec{k}) \quad f \text{ is some function to be summed}$$

$$= \left(\frac{L}{2\pi}\right)^3 \sum_{n_x=-\infty}^{\infty} \sum_{n_y=-\infty}^{\infty} \sum_{n_z=-\infty}^{\infty} f(\vec{k}) \frac{2\pi}{L} \times \frac{2\pi}{L} \times \frac{2\pi}{L}$$

$$= \left(\frac{L}{2\pi}\right)^3 \sum_{k_x=-\infty}^{\infty} \sum_{k_y=-\infty}^{\infty} \sum_{k_z=-\infty}^{\infty} f(\vec{k}) \Delta k_x \Delta k_y \Delta k_z$$

$$= \frac{V}{(2\pi)^3} \int_{-\infty}^{\infty} dk_x \int_{-\infty}^{\infty} dk_y \int_{-\infty}^{\infty} dk_z f(\vec{k})$$

$$\Rightarrow \boxed{\sum_{\vec{k}} \xrightarrow{L \rightarrow \infty} \frac{V}{(2\pi)^3} \int d^3 k}$$

Calculation of E_F : $N = \sum_{\vec{k}} n_{\vec{k}} = \sum_{\vec{k}} \frac{g}{e^{\beta(E_{\vec{k}} - \mu)} + 1} = \sum_{|\vec{k}| < k_F} g$

where k_F is given by $E_F = \frac{\hbar^2 k_F^2}{2m}$
and g is the spin degeneracy, $g = 2s + 1 = 2$ for electrons.

$$N = \frac{V}{(2\pi)^3} \int_{|\vec{k}| < k_F} d^3 k \cdot 2 = \frac{2V}{(2\pi)^3} \frac{4\pi}{3} k_F^3 = \frac{V k_F^3}{3\pi^2}$$

$$\Rightarrow k_F = (3\pi^2 n)^{1/3}$$

$$\Rightarrow E_F = \frac{\hbar^2}{2m} (3\pi^2 n)^{2/3}$$

The internal energy of the electron gas is

$$\begin{aligned} U &= \sum_{\vec{k}} n_{\vec{k}} E_{\vec{k}} = \sum_{\vec{k}} \frac{g E_{\vec{k}}}{e^{\beta(E_{\vec{k}} - \mu)} + 1} = \frac{2V}{(2\pi)^3} \int_{|\vec{k}| < k_F} d^3 k \frac{\hbar^2 k^2}{2m} \\ &= \frac{V \hbar^2}{(2\pi)^3 m} \int_0^{k_F} 4\pi k^2 dk \cdot k^2 = \frac{\hbar^2 V}{2\pi^2 m} \frac{k_F^5}{5} = \frac{V E_F}{5\pi^2} k_F^3 = \frac{V 3\pi^2 n}{5\pi^2} E_F = \frac{3N}{5} E_F \end{aligned}$$

$$\Rightarrow U = \frac{3}{5} N E_F$$

The Fermi temperature is defined by $k_B T_F = E_F$.

If the temperature $T \ll T_F$, the Fermi gas becomes strongly degenerate.

$e^{\beta\mu} = D$ is known as the **degeneracy parameter**.

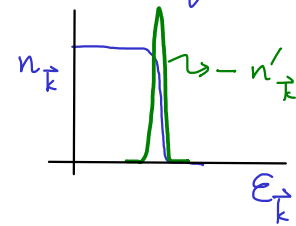
The crossover from degeneracy to non-degeneracy happens around $D = \exp(1)$.

Electronic contribution to the specific heat of metals

At a small non-zero temperature, $T \ll T_F$, the Fermi distribution function is almost a step function around $\mu \approx E_F$.

$$\begin{aligned}
 N &= \frac{V}{(2\pi)^3} \int_0^\infty 4\pi k^2 dk \frac{2}{e^{\beta(\frac{\hbar^2 k^2}{2m} - \mu)} + 1} \\
 &= \frac{V}{\pi^2} \int_0^\infty dk \left(\frac{1}{e^{\beta(\frac{\hbar^2 k^2}{2m} - \mu)} + 1} \right) k^2 \\
 &= \frac{V}{\pi^2} \left[\underbrace{\left(\frac{1}{e^{\beta(\frac{\hbar^2 k^2}{2m} - \mu)} + 1} \right)}_0 k^3 / 3 \right]_0^\infty - \int_0^\infty dk \left(\frac{1}{e^{\beta(\frac{\hbar^2 k^2}{2m} - \mu)} + 1} \right)' \frac{k^3}{3}
 \end{aligned}$$

If $T \ll T_F$, the derivative of the Fermi distribution function is sharply peaked near k_F .



$$-\left(\frac{1}{e^{\beta(\frac{\hbar^2 k^2}{2m} - \mu)} + 1} \right)' = \frac{e^{\beta(\frac{\hbar^2 k^2}{2m} - \mu)} \beta \frac{\hbar^2 k}{m}}{\left(e^{\beta(\frac{\hbar^2 k^2}{2m} - \mu)} + 1 \right)^2}$$

$$\Rightarrow N \approx \frac{V}{6\pi^2} \frac{\beta \hbar^2}{m} \int_0^\infty \frac{e^{\beta(\frac{\hbar^2 k^2}{2m} - \mu)} k^4}{\left(e^{\beta(\frac{\hbar^2 k^2}{2m} - \mu)} + 1 \right)^2} dk$$

$$\text{Let } \beta \left(\frac{\hbar^2 k^2}{2m} - \mu \right) = x \Rightarrow \frac{\beta \hbar^2}{m} k dk = dx$$

$$k^2 = \frac{2m}{\beta \hbar^2} (x + \beta \mu)$$

$$k \rightarrow 0 \Rightarrow x \rightarrow -\beta \mu$$

$$k \rightarrow \infty \Rightarrow x \rightarrow \infty$$

$$k \approx k_F \Rightarrow |x| \ll 1$$

$$\Rightarrow N \approx \frac{V}{6\pi^2} \int_{-\beta \mu}^\infty \frac{e^x}{(e^x + 1)^2} \left(\frac{2m}{\beta \hbar^2} \right)^{3/2} (x + \beta \mu)^{3/2} dx$$

$$\approx \frac{V}{6\pi^2} \left(\frac{2m\mu}{\hbar^2} \right)^{3/2} \int_{-\infty}^\infty \frac{e^x}{(e^x + 1)^2} \left(1 + \frac{3}{2} \frac{x}{\beta \mu} + \frac{3}{8} \left(\frac{x}{\beta \mu} \right)^2 + \dots \right) dx$$

Since the integrand is sharply peaked near $x=0$, we have replaced the lower limit with $-\infty$.

$\frac{e^x}{(e^x + 1)^2}$ is an even function of x

$$\Rightarrow N \approx \frac{V}{6\pi^2} \left(\frac{2m\mu}{\hbar^2} \right)^{3/2} 2 \int_0^\infty \frac{e^x}{(e^x + 1)^2} \left(1 + \frac{3}{8} \left(\frac{x}{\beta \mu} \right)^2 + \dots \right) dx$$

$$\int_0^{\infty} \frac{e^x}{(e^x + 1)^2} dx$$

$$= \int_0^{\infty} \frac{dz}{z^2} = \frac{1}{2}$$

$$\text{and } \int_0^{\infty} \frac{x^2 e^x}{(e^x + 1)^2} dx = \frac{\pi^2}{6}$$

$$e^x + 1 = z \Rightarrow e^x dx = dz$$

$$x \rightarrow -\beta\mu \Rightarrow z \rightarrow e^{-\beta\mu} + 1, \quad x \rightarrow \infty \Rightarrow z \rightarrow \infty$$

$$\Rightarrow N \approx \frac{V}{3\pi^2} \left(\frac{2m\mu}{\hbar^2} \right)^{3/2} \left[1 + \frac{\pi^2}{8} (\beta\mu)^{-2} \right]$$

$$\beta\mu \approx \frac{T_F}{T} \Rightarrow N \approx \frac{V}{3\pi^2} \left(\frac{2m\mu}{\hbar^2} \right)^{3/2} \left[1 + \frac{\pi^2}{8} \left(\frac{T}{T_F} \right)^2 \right]$$

$$\Rightarrow \mu \approx \left(3\pi^2 n \right)^{2/3} \frac{\hbar^2}{2m} \left[1 + \frac{\pi^2}{8} \left(\frac{T}{T_F} \right)^2 \right]^{-2/3}$$

$$\mu \approx E_F \left[1 - \frac{\pi^2}{12} \left(\frac{T}{T_F} \right)^2 \right] \quad \text{---- (1) Sommerfeld Equation}$$